



Maintenance Schedule

Diesel Engine

12V/16V/18V2000Bx6

12V/16V/18V2000Bx6F

12V/16V/18V2000Gx6

Application Group 3A, 3B, 3E, 3F, 3G

MS50173/04E

Engine model	kW/cyl.	rpm	Application group
12V2000B26F	55 kW/cyl.	1500	3A, heavy duty, unrestricted, ICXN
12V2000B76 switchable	59/60 kW/cyl.	1500/1800	3B, continuous service, variable load, ICXN
16V2000B26F	44 kW/cyl.	1500	3A, heavy duty, unrestricted, ICXN
16V2000B76 switchable	56/62 kW/cyl.	1500/1800	3B, continuous service, variable load, ICXN
18V2000B26F	49 kW/cyl.	1500	3A, heavy duty, unrestricted, ICXN
18V2000B76 switchable	61/61 kW/cyl.	1500/1800	3B, continuous service, variable load, ICXN
12V2000G16F	55 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN
12V2000G26F	59 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN
16V2000G16F	50 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN
16V2000G26F	56 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN 3F, heavy duty for DCP, unrestricted, ICXN 3G, heavy duty, intermittent, ICXN
16V2000G36F	63 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN
18V2000G26F	61 kW/cyl.	1500	3B, continuous service, variable load, ICXN 3E, emergency service, ICXN 3F, heavy duty for DCP, unrestricted, ICXN 3G, heavy duty, intermittent, ICXN

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product manufactured by Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of operational availability. The maintenance intervals as well as the required maintenance task scopes are based on operational experience and are therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of demanding operational and ambient conditions.

Observance of the specified maintenance intervals is essential to maintain product safety.

The intervals at which the maintenance tasks are to be performed are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule)

QL3: Maintenance work which requires partial disassembly of the product

QL4: Maintenance work which requires complete disassembly of the product

Additional notes on maintenance and preservation:

Maintenance activities stipulated in any documentation provided by third-parties (suppliers and component manufacturers), particularly those relating to individual components and subsystems, may differ from the information specified in this Maintenance Schedule. The Maintenance Schedule provided by Rolls-Royce Solutions shall take precedence in such case.

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

Refer to the applicable Fluids and Lubricants Specifications for details of fluid and lubricant change intervals.

If the product is to remain out of service for a longer period of time, Rolls-Royce Solutions recommends carrying out a monthly test run until the engine has reached its normal operating temperature. If the engine is scheduled to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

TM-ID: 000004281 - 014

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix application group 3A, 3B, 3E, 3F, 3G

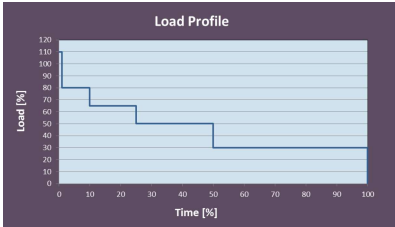
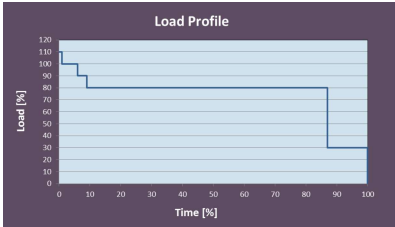
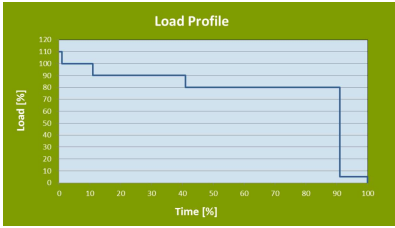
Load profile		Load factor %	Load indicator %	12V2000B26F 750 kVA 16V2000B26F 800 kVA 18V2000B26F 1000 kVA	12V2000G16F 750 kVA 12V2000G26F 800 kVA 16V2000G16F 910 kVA 16V2000G26F 1000 kVA	16V2000G36F 1135 kVA emissions-optimized 18V2000G26F 1250 kVA 12V2000B76 switchable 18V2000B76 switchable	16V2000G36F 1135 kVA be-optimized 16V2000B76 switchable	
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	1	≤ 46	≤ 5	24,000	22,000	22,000	18,000	
100 ₍₉₀₋₁₀₀₎	—							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	9							
65 ₍₅₀₋₆₅₎	15							
50 ₍₃₀₋₅₀₎	25							
30 ₍₅₋₃₀₎	50							
Idle speed ₍₀₋₅₎	—	≤ 75	≤ 18	24,000	18,000	18,000	12,000	
110 ₍₁₀₀₋₁₁₀₎	1							
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	3							
80 ₍₆₅₋₈₀₎	78							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	13							
Idle speed ₍₀₋₅₎	—	≤ 80	≤ 32	22,000	16,000	16,000	10,000	Similar application group 3B 
110 ₍₁₀₀₋₁₁₀₎	1							
100 ₍₉₀₋₁₀₀₎	10							
90 ₍₈₀₋₉₀₎	30							
80 ₍₆₅₋₈₀₎	50							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle speed ₍₀₋₅₎	9							

Table 2: Allocation matrix application group 3A, 3B, 3E, 3F, 3G

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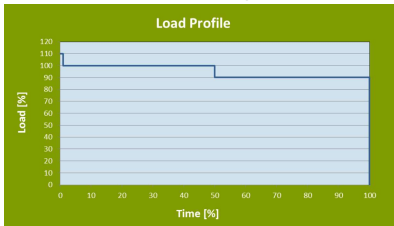
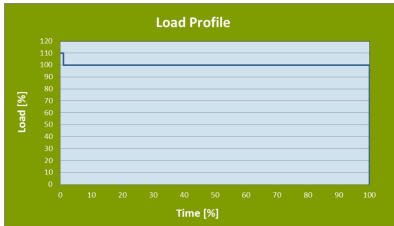
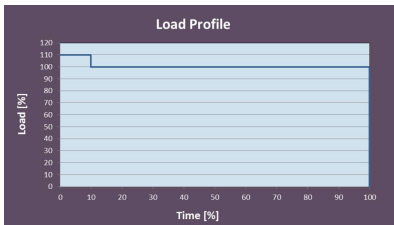
Load profile		Load factor %	Load indicator %	12V2000B26F 750 kVA 16V2000B26F 800 kVA 18V2000B26F 1000 kVA	12V2000G16F 750 kVA 12V2000G26F 800 kVA 16V2000G16F 910 kVA 16V2000G26F 1000 kVA	16V2000G36F 1135 kVA emissions-optimized 18V2000G26F 1250 kVA 12V2000B76 switchable 18V2000B76 switchable	16V2000G36F 1135 kVA be-optimized 16V2000B76 switchable	
Load %	Time %							
110 (100-110)	1	≤ 95	≤ 70	20,000	10,000	10,000	7,000	Similar application group 3E Load Profile 
100 (90-100)	49							
90 (80-90)	50							
80 (65-80)	—							
65 (50-65)	—							
50 (30-50)	—							
30 (5-30)	—							
Idle speed (0-5)	—							
110 (100-110)	1	≤ 100	≤ 102	18,000	8,000	8,000	5,000	Similar application group 3A Similar application group 3F Similar application group 3G Load Profile 
100 (90-100)	99							
90 (80-90)	—							
80 (65-80)	—							
65 (50-65)	—							
50 (30-50)	—							
30 (5-30)	—							
Idle speed (0-5)	—							
110 (100-110)	10*	≤ 101	≤ 116	16,000	7,000	7,000	4,000	Load Profile 
100 (90-100)	90							
90 (80-90)	—							
80 (65-80)	—							
65 (50-65)	—							
50 (30-50)	—							
30 (5-30)	—							
Idle speed (0-5)	—							
*No continuous operation at 110% power								

Table 3: Allocation matrix application group 3A, 3B, 3E, 3F, 3G

1.4 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
0020	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.	WM00980	1000	0	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.5 Cyclical maintenance tasks 24000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check engine oil level		WM00285
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Visually inspect engine for leaks and check its general condition		WM00286
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check maintenance indicator of air filter		WM00287
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check relief bore of coolant pump		WM00288
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	When engine is running, check for abnormal running noises, exhaust gas color and vibrations		WM00289
QL1	-	1 MON	ENGINE OPERATIONAL MONITORING	Perform test run, minimum duration: until operating temperature is reached, no less than 1/3 load		WM00296
QL1	-	1 YR	ENGINE MOUNTING	Measure height of rubber element		WM00126
QL1	-	2 YR	LUBE OIL SYSTEM	Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest		WM00067
QL1	-	18 YR	HP FUEL LINE HP LINE	Replace HP line		WM00172
QL1	500	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace easy-change fuel filter		WM00086
QL1	1000	1 YR	ENGINE MOUNTING	Check general condition of resilient mounts (visual inspection)		WM00050
QL1	1000	1 YR	ENGINE MOUNTING	Check securing screws for firm seating		WM00136
QL1	1000	2 YR	PRELUBRICATION	Visually inspect pump	X	WM00069
QL1	1000	2 YR	AUXILIARY POWER TAKE-OFF	Check belt condition, replace if necessary. Adjust tension.	X	WM00170
QL1	2000	18 YR	VALVE GEAR	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		WM00094
QL1	3000	3 YR	COOLANT SYSTEM	Replace valve cover	X	WM00664
QL1	4000	3 YR	AIR SUPPLY SYSTEM	Replace air filter		WM00076
QL1	8000	3 YR	CRANKCASE VENTILATION	Replace oil separator (impactor)		WM00106
QL1	8000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace fuel injection valve/injector		WM00090
QL1	8000	18 YR	RUNNING GEAR	Inspect cylinder liner with endoscope		WM00110
QL1	8000	18 YR	GOVERNOR	Reset parameters for drift correction (CDC) and enter injector coding (IIG)		WM00154
QL1	8000	18 YR	PRESSURE PIPE NECK	Replace pressure pipe neck in cylinder head		WM00186
QL3	12000	6 YR	HOSE	Replace hose line		WM00006
QL3	12000	6 YR	RUBBER SLEEVE	Replace rubber sleeve		WM00008
QL3	12000	18 YR	COOLANT SYSTEM	Check HT thermostat housing and replace thermal actuator		WM00056
QL3	12000	18 YR	CYLINDER HEAD	Visually inspect piston crown		WM00101
QL3	12000	18 YR	CYLINDER HEAD	Cylinder liners, visually inspect running pattern		WM00109
QL3	12000	18 YR	EXHAUST TURBOCHARGER	Replace exhaust turbocharger		WM00117
QL3	12000	18 YR	AIR PIPING TO CYLINDERS	Clean air piping		WM00164
QL3	12000	18 YR	COOLANT SYSTEM	Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)		WM00921
QL3	12000	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in HT circuit		WM00974
QL3	12000	18 YR	GENERATOR	Overhaul or replace battery-charging generator	X	WM00976
QL3	12000	18 YR	CYLINDER HEAD	Overhaul or replace cylinder head		WM00980
QL3	12000	18 YR	LUBE OIL SYSTEM	Clean oil cooler and check for leaks or replace		WM00981
DAY = Days MON = Months YR = Years HRS = Hours						

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Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL3	24000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP fuel sensor		WM00011
QL3	24000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace pressure relief valve		WM00088
QL3	24000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP pump		WM00091
QL3	24000	18 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace wiring harness		WM00137
QL3	24000	18 YR	AUXILIARY POWER TAKE-OFF	Replace bearing	X	WM00162
QL3	24000	18 YR	VALVE GEAR	Check rocker arm, valve bridge, pushrods and ball joints for wear		WM00166
QL3	24000	18 YR	RUNNING GEAR	Replace vibration damper		WM00439
QL3	24000	18 YR	STARTING SYSTEM	Overhaul or replace starter		WM00971
QL3	24000	18 YR	PRELUBRICATION	Overhaul or replace oil priming pump		WM00978
QL3	24000	18 YR	ENGINE MOUNTING	Replace resilient mounts of engine		WM01009
QL4	24000	18 YR	ENGINE, GENERAL	Replace elastomeric members and seals		WM00001
QL4	24000	18 YR	ENGINE, GENERAL	Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary		WM00002
QL4	24000	18 YR	LUBE OIL SYSTEM	Check pump, replace if necessary		WM00070
QL4	24000	18 YR	RUNNING GEAR	Replace conrod bearings		WM00099
QL4	24000	18 YR	RUNNING GEAR	Replace rings		WM00100
QL4	24000	18 YR	RUNNING GEAR	Replace crankshaft bearings		WM00103
QL4	24000	18 YR	GEAR TRAIN	Check tooth flanks in gear train for wear (visual inspection), replace bushing		WM00104
QL4	24000	18 YR	RUNNING GEAR	Replace cylinder liners		WM00108
QL4	24000	18 YR	CAMSHAFT	Replace camshaft bearings		WM00111
QL4	24000	18 YR	RUNNING GEAR	Check conrod, replace as necessary		WM00451
Preheater						
QL1	-	1 YR	PREHEATING	Preheater, check for function and leaks	X	WM00222
QL1	-	3 YR	PREHEATING	Preheater, replace thermostat (electric)	X	WM00148
QL3	24000	18 YR	PREHEATING	Overhaul or replace preheating unit	X	WM00979
Coolant cooler						
QL1	500	6 MON	COOLING SYSTEM	Check coolant cooler externally for contamination and leaks	X	WM00229
Fuel cooler						
QL1	500	6 MON	FUEL COOLER	Check cooler elements externally for contamination and leaks	X	WM00228
Generator						
QL1	-	1 YR	GENERATOR MOUNTING	Measure height of rubber element of generator mounts	X	WM00190
QL1	1000	1 YR	GENERATOR MOUNTING	Check securing screws of generator mounts for secure seating	X	WM00188
QL1	1000	1 YR	GENERATOR MOUNTING	Check general condition of rubber elements of generator mounts (visual inspection)	X	WM00189
QL3	24000	18 YR	GENERATOR MOUNTING	Replace resilient mounts of generator	X	WM01010
Fuel prefilter						
QL1	-	1 MON	FUEL SYSTEM (LOW PRESSURE)	Drain water and contaminants from fuel prefilter	X	WM00215
QL1	-	1 MON	FUEL SYSTEM (LOW PRESSURE)	Check position of contamination indicator of fuel prefilter	X	WM00216
QL1	500	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace filter element	X	WM00214
DAY = Days MON = Months YR = Years HRS = Hours						

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